

Syllabus

SURV 626 Sampling I

2 credits/ 4 ECTS

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September 25 – November 13, 2025

Short Course Description

Sampling is an applied statistics methods course but differs from most statistics courses because it is concerned almost exclusively with the design of data collection. Little of the analysis of collected data will be discussed in the course. The course will concentrate on problems of applying sampling methods to human populations, since sampling human populations poses a number of particular problems not found in sampling of other types of units. The principles of sample selection, though, can be applied to many other types of populations.

Course Objectives

By the end of the course, students will...

- *understand the basic ideas, concepts and principles of probability sampling from an applied perspective*
- *be able to identify and appropriately apply sampling techniques to survey design problems*
- *be able to compute the sample size for a variety of sample designs*
- *understand and be able to assess the impact of the sample design on survey estimates*
- *be able to estimate the precision of the survey statistics using different estimation techniques*

Prerequisites

The course is presented at an intermediate statistical level. While we will not develop mathematical aspects of sampling theory, statistical notation and outlines of some algebraic proofs will be given. A sound background in applied statistics, proficiency in mathematics, including basic algebra, is necessary, since some algebraic derivations will be presented (although little emphasis will be placed on the derivations). A thorough understanding of the notation and algebraic results will be required.

Class Structure and Course Concept

This is an online course using a flipped classroom design. It covers the same material and content as an on-site course but runs differently. In this course, you are responsible for watching video-recorded lectures and reading the required literature for each unit prior to participating in mandatory weekly one-hour online meetings where students have the chance to discuss the materials from a unit with the instructor.

Although this is an online course where students have more freedom in when they engage with the course materials, students are expected to spend the same amount of time overall on all activities in the course – including preparatory activities (readings, studying), in-class-activities (watching prerecorded videos, attending the live online meetings), and follow-up activities (working on assignments and exams) – as in an on-site course. As a rule of thumb, you can expect to spend approximately 3h/week on in-class activities and 9 hours per week on out-of-class activities (preparing for class, readings, assignments, projects, studying for quizzes and exams). Therefore, the workload for all courses will be approximately 12h/week. Please note that the actual workload will depend on your personal knowledge.

Mandatory Weekly Online Meetings

Thursdays 12:00 PM ET / 6:00 PM CET, starting September 25, 2025

Meetings will be held online through Zoom. Follow the link to the meeting sessions on the course website on <https://umd.zoom.us>. If video participation via Internet is not possible, arrangements can be made for students to dial in and join the meetings via telephone.

In preparation for the weekly online meetings, students are expected to watch the lecture videos and read the assigned literature before the start of the meeting. In addition, students are encouraged to post questions about the materials covered in the videos and readings of the week in the forum before the meetings (deadline for posting questions is Wednesday, 8:00 PM ET / 2:00 AM CET).

Students have the opportunity to use the Conferences feature in Canvas to connect with peers outside the scheduled weekly online meetings (e.g., for study groups). Students are not required to use Canvas Conferences and can of course use other online meeting platforms such as Google Hangouts, Skype or Microsoft Teams.

Daylight savings time ends in Europe on October 26, 2025, and clocks are turned back 1 hour.

Daylight savings time ends in the USA on November 2, 2025.

Therefore, look carefully at the times of meetings and deadlines!

Grading

Grading will be based on:

- Homework assignments (50% of the grade)
- Quizzes (15% of the grade)
- Participation in discussion during the weekly online meetings, submission of questions to the weekly discussion forum (deadline: Wednesday, 8:00 PM ET / 2:00 AM CET), and demonstrating understanding of the required readings and video lectures, and positive contributions in the discussion forum (10% of grade)
- A final open-book online exam (25% of grade)

A+ 100 - 97

A 96 - 93

A- 92 - 90

B+ 89 - 87

B 86 - 83

B- 82 - 80

Etc.

The grading scale is a base scale recommended by the MDM. Variations for grading on a scale are at the discretion of the instructor.

The final grade will be communicated under the assignment "Final Grade" in the Canvas course. Please note that the letter grade written in parentheses in Canvas is the correct final grade. The point-grade displayed alongside the letter grade is irrelevant and can be ignored. Dates of when assignment will be due are indicated in the syllabus. Extensions will be granted sparingly and are at the instructor's discretion.

Homework assignments

The homework assignments will involve small-scale, sample design problems that will require you to identify and apply the methods and techniques covered in the lectures and assigned readings. The questions will require mathematical calculations, and you will be asked to select samples using different sampling schemes. Although some examples of statistical software will be provided, none of the homework problems will require their use, and the assignments should preferably be solved by hand, with a calculator, or in a spreadsheet, so that you can have a more robust understanding of the concepts being applied in these exercises. Use the homework assignments as an indicator of your progress in this course.

Homework solutions should be submitted electronically via the course web site Assignment tool as an attachment. You must submit solutions, handwritten or typed, in a single PDF format file, with name and homework set number at the top of the first page, and page numbers at the bottom of each page. Handwritten versions must be fully legible: if the instructor cannot read the homework, it will be returned ungraded. Files must be submitted in a standard name convention: 'Surname First Initial HW#.pdf'. For example, 'Nishimura R HW1.pdf'. Homework problems will be graded on a 100-point scale. The submitted homework will be marked electronically and returned via the Assignment tool, along with a copy of the homework solution.

Homework assignments are due the Tuesday after the online meeting (see schedule syllabus below). Late homework will not be accepted, except in case of emergencies, which should be reported to the instructor in advance through a request made in writing by email no less than 24 hours before the homework is due, and a reason must be given for the need to submit late. Late homework submission permission is not guaranteed.

Study groups are encouraged. However, group answers are not acceptable, and each student must submit individual homework solutions. You are encouraged to ask and answer questions in the discussion forum about the homework assignments, but you should not request for or provide entire solutions. If this behavior is detected, there will be a 50% penalty on your grade for that assignment.

Quizzes

During the first five minutes of each class session, there will be a closed book, closed notes quiz with three to five multiple choice questions about the assigned readings for that week (see textbooks and assigned readings and syllabus schedule). The questions will not involve any mathematical calculation and will assess the student's understanding of some of the basic concepts and ideas of the content covered on the assigned readings, which will not necessarily be covered in the lectures. The students are encouraged to ask questions in the discussion forum about the assigned readings. There will be no make-up quizzes, but we will drop the two lowest quizzes before calculating the final grade.

Class participation

In preparation for the weekly online meetings, students are expected to watch the lecture videos and read the assigned literature before the start of the meeting. Please be prepared to contribute to the class discussion: everyone is expected to contribute. In addition, students are encouraged to post questions

about the materials covered in the videos and readings of the week in the weekly discussion before the meetings (deadline for posting questions is Wednesday, 8:00 PM ET / 2:00 AM CET).

Participation through the discussion forum, either by asking or answering questions, is encouraged and positive contributions will be rewarded on your final grade. However, you should not request for or provide entire homework solutions. If this behavior is detected, there will be a 50% penalty on your grade for that assignment.

Final open-book exam

The final cumulative, open-book, take-home exam will be available on the course website from November 20, 12:00 PM ET / 6:00 PM CEST to November 27, 12:00 PM ET / 6:00 PM CET. Students will have 48 hours to complete it starting from the time the exam is opened on the course website. The solution of the exam should be uploaded to the course. If the student is unable to take the exam on the scheduled week due to prior commitments, they should contact the instructor as soon as possible to make special arrangements.

Dates of when assignments will be due are indicated in the syllabus.

Technical Equipment Needs

The learning experience in this course will mainly rely on the online interaction between the students and the instructors during the weekly online meetings. Therefore, we encourage all students in this course to use a web camera and a headset. Decent quality headsets and web cams are available for less than \$20 each. We ask students to refrain from using built-in web cams and speakers on their desktops or laptops. We know from our experience in previous online courses that this will reduce the quality of video and audio transmission and therefore will decrease the overall learning experience for all students in the course. In addition, we suggest that students use a wire connection (LAN), if available, when connecting to the online meetings. Wireless connections (WLAN) are usually less stable and might be dropped.

Mannheim Business School would also like to officially inform you that, in order to facilitate your participation in this course, your personal data will be processed by and on systems run by MBS and our subcontractors. You can find detailed information in our privacy policy and information for data subjects [here](#).

Long Course Description

In most statistics courses, students learn numerous data analysis methods and techniques, especially focused on how to make inferences to a population based on stochastic models. Sampling is a unique course in statistics for two reasons: (i) it is concerned almost exclusively with the sample design, and (ii) it adopts a different inferential approach, which does not rely on assumptions about the stochastic distribution of the study variables, but rather on the probability distribution produced by the randomization process on the selection of a probability sample. Therefore, an applied sampling course provides students with essential exposure to a different perspective in statistics, which they might not have had contact with other courses.

As this class is likely the first contact with the subject that most students have had, the primary learning goal we have for students in this course is to understand the basic ideas, concepts and principles underlying sampling methods. The role of randomization in the sample selection and in making inferences to the population, the abstract idea of all possible samples that could have been selected from a population for a given sampling design, and the difference between sampling error and bias are examples of such principles, concepts and ideas.

We also expect that students will acquire some of this basic knowledge through a list of assigned reading we have prepared for this course. Most of them are from the textbook, *Survey Sampling*, by Leslie Kish, which, though written in the 1960s, remains an excellent applied reference. We made this choice because most, if not all, of the more recent books on sampling focus on the statistical theory but lack a more applied perspective as is presented in Kish. However, many other sampling techniques have been developed since Leslie Kish *Survey Sampling*. For that reason, we also assign more recent research papers about selected topics that are either not covered in the book or that we believe are better developed in other references.

As another learning goal of this course, students should be able to identify and apply appropriate sampling techniques across a range of different methods covered in class (simple random sampling, stratification, cluster sampling, multistage sampling, systematic sampling and probability proportional to size) in sample design problems. As Aristotle once said: "For the things we have to learn before we can do, we learn by doing". We believe this is especially true for statistics: students learn much more by doing exercises. For that reason, we use homework assignments as our main teaching method and assessment for this goal. We carefully design these assignments, so students apply what they have learned in lectures and in the readings. Based on the students' performance, we determine which topics need to be reinforced.

Readings

Mandatory Readings

Survey Sampling by Leslie Kish (John Wiley and Sons, Inc., New York, 1965)

OR

Introduction to Survey Sampling by Graham Kalton (Sage Publications, Beverly Hills, 2020)

Complementary Readings

[1] Rust, K. (1985). Variance estimation for complex estimators in sample surveys. *Journal of Official Statistics*, 1(4), pp. 381-397.

[2] Kish, L. and Frankel, M. (1974). Inference from complex samples," *Journal of the Royal Statistical Society, Series B*, 36, pp. 1 - 37.

Interested students may find the following recommended books helpful in preparing for the course and as supplemental reading to several lecture topics: *Sample Survey Methods and Theory, Volume 1*, by Morris Hansen, et al. (New York: John Wiley and Sons, Inc., 1953), and *Sampling Techniques*, 3rd edition, by William G. Cochran (New York: John Wiley and Sons, Inc., 1977).

Academic Conduct

Clear definitions of the forms of academic misconduct, including cheating and plagiarism, as well as information about disciplinary sanctions for academic misconduct may be found at

<https://www.president.umd.edu/sites/president.umd.edu/files/documents/policies/III-100A.pdf> (University of Maryland)

Knowledge of these rules is the responsibility of the student and ignorance of them does not excuse misconduct. The student is expected to be familiar with these guidelines before submitting any written work or taking any exams in this course. Lack of familiarity with these rules in no way constitutes an

excuse for acts of misconduct. Charges of plagiarism and other forms of academic misconduct will be dealt with very seriously and may result in oral or written reprimands, a lower or failing grade on the assignment, a lower or failing grade for the course, suspension, and/or, in some cases, expulsion from the university.

Accommodations for Students with Disabilities

In order to receive services, students at the University of Maryland must contact the Accessibility & Disability Service (ADS) office to register in person for services. Please call the office to set up an appointment to register with an ADS counselor. Contact the ADS office at 301.314.7682; <https://www.counseling.umd.edu/ads/>.

Course Evaluation

In an effort to improve the learning experience for students in our online courses, students will be invited to participate in an online course evaluation at the end of the course. Participation is entirely voluntary and highly appreciated.

Sessions

Week 1: Introduction; Course Perspectives

Video lecture: available Thursday, September 18, 2025

Design

Typology and vocabulary

Notation

Sampling distribution

Online meeting: Thursday, September 25, 2025, 12:00 PM EDT / 6:00 PM CEST

Homework Assignment 1: due Tuesday, September 30, 2025, 12:00 PM EDT / 6:00 PM CEST

Mock Quiz: During the first five minutes of each class session, there will be a closed book, closed notes quiz with three to five multiple choice questions about the assigned readings for that week.

Required Readings:

Kish (1965), Chapter 1.

OR

Kalton (2020), Chapter 1.

Week 2: Simple Random Sampling

Video lecture: available Thursday, September 25, 2025

Simple Random Sampling

Online meeting: Thursday, October 2, 2025, 12:00 PM EDT / 6:00 PM CEST

Homework Assignment 2: due Tuesday, October 7, 2025, 12:00 PM EDT / 6:00 PM CEST

Online Quiz 1: During the first five minutes of each class session, there will be a closed book, closed notes quiz with three to five multiple choice questions about the assigned readings for that week.

Required Readings:

Kish (1965), Sections 2.1-2.6.

OR

Kalton (2020), Chapters 2.

Optional Videos:

Three Frame Problems

Frames: Clusters

Optional Readings:

Kish (1965), Sections 2.7, 11.1-11.3.

OR

Kalton (2020), Chapter 8.

Week 3: Stratified Sampling

Video lecture: available Thursday, October 2, 2025

Stratified Random Sampling

Stratified Sampling Allocations

Online meeting: Thursday, October 9, 2025, 12:00 PM EDT / 6:00 PM CEST

Homework Assignment 3: due Tuesday, October 14, 2025, 12:00 PM EDT / **5:00 PM CEST**

Online Quiz 2: During the first five minutes of each class session, there will be a closed book, closed notes quiz with three to five multiple choice questions about the assigned readings for that week.

Required Readings:

Kish (1965), Sections 3.1-3.5.

OR

Kalton (2020), Chapter 4.

Optional Videos:

Stratification Topics

Optional Readings:

Kish (1965), Sections 3.6, 4.5A.

Week 4: Systematic selection

Video lecture: available Thursday, October 9, 2025

Systematic Selection

Online meeting: Thursday, October 16, 2025, 12:00 PM EDT / **5:00 PM CET**

Homework Assignment 4: due Tuesday, October 21, 2025, 12:00 PM ET / 6:00 PM CET

Online Quiz 3: During the first five minutes of each class session, there will be a closed book, closed notes quiz with three to five multiple choice questions about the assigned readings for that week.

Required Readings:

Kish (1965), 4.1-4.3.

OR

Kalton (2020), Chapters 3.

Week 5: Cluster Sampling

Video lecture: available Thursday, October 16, 2025

Cluster Sampling

Two-stage Cluster Sampling

Cluster Sampling Subsample Size

Online meeting: Thursday, October 23, 2025, 12:00 PM ET / 6:00 PM CET

Homework Assignment 5: due Tuesday, October 28, 2025, 12:00 PM ET / 6:00 PM CET

Online Quiz 4: During the first five minutes of each class session, there will be a closed book, closed notes quiz with three to five multiple choice questions about the assigned readings for that week.

Required Readings:

Kish (1965), Sections 5.1-5.4, 8.3.

OR

Kalton (2020), Chapter 5.

Week 6: Unequal-Sized Clusters I

Video lecture: available Thursday, October 23, 2025

Unequal Sized Cluster Sampling

Stratified Unequal Sized Clusters

Please keep in mind that winter time begins in Europe on October 26, 2025 and clocks are turned back 1 hour. Winter time begins in the USA on November 2, 2025. Therefore, look carefully at the times of meetings and deadlines!

Online meeting: Thursday, October 30, 2025, 12:00 PM ET / 6:00 PM CET

Homework Assignment 6: due Tuesday, November 4, 2025, 12:00 PM ET / 6:00 PM CET

Online Quiz 5: During the first five minutes of each class session, there will be a closed book, closed notes quiz with three to five multiple choice questions about the assigned readings for that week.

Required Readings:

Kish (1965), Sections 6.1-6.5.

OR

Kalton (2020), Chapter 5.

Week 7: Unequal-Sized Clusters II

Video lecture: available Thursday, October 30, 2025

Complex Designs

Online meeting: Thursday, November 6, 2025, 12:00 PM ET / 6:00 PM CET

Homework Assignment 7: due Tuesday, November 11, 2025, 12:00 PM ET / 6:00 PM CET

Online Quiz 6: During the first five minutes of each class session, there will be a closed book, closed notes quiz with three to five multiple choice questions about the assigned readings for that week.

Required Readings:

Kish (1965), Sections 6.1-6.5.

OR

Kalton (2020), Chapter 5.

Week 8: Unequal-Sized Clusters III

Video lecture: available Thursday, November 6, 2025

PPS Sampling

PPeS Sampling

Online meeting: Thursday, November 13, 2025, 12:00 PM ET / 6:00 PM CET

Homework Assignment 8: due Tuesday, November 18, 2025, 12:00 PM ET / 6:00 PM CET

Online Quiz 7: During the first five minutes of each class session, there will be a closed book, closed notes quiz with three to five multiple choice questions about the assigned readings for that week.

Required Readings:

Kish (1965), Sections 7.1-7.5.

OR

Kalton (2020), Chapter 6.

Optional Videos:

Forming Computing Units

Taylor Series Expansion

Replicated Sampling

Balanced Repeated Replication

Jackknife Replication

Required Readings:

Kish (1965), Sections 8.6, 4.4, 14.1-14.2, 13.1-13.3 .

OR

Kalton (2020), Chapter 12.

[1] Rust (1985)

[2] Kish and Frankel (1974)

Final exam

Final exam opens Thursday, November 20, 2025, 12:00 PM ET/6:00 PM CET

Final exam closes Thursday, November 27, 2025, 12:00 PM ET/6:00 PM CET